

Software Requirements Specification for Race Walking Data Visualizer

Version 1.0

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1. Document History

Names	Date	Reason	Version
Elvis Cao, Ostap Bezrukyy, Jiacong Li, Sotero Casiano, Nico Saputra, Zach Bridgers, Jay Dave	1/13/2025	Initial Draft	1.0

2. Introduction

2.1 Purpose

This document provides the requirement specifications for the Race Walking Data Visualizer application. It specifies user interface attributes, functional and nonfunctional requirements, and longterm ideas for the evolution of the system.

2.2 Overview

The Race Walking Data Visualizer is an application designed to streamline the analysis of race walking events. It provides automated report generation and creates visualizations of athletes' infraction patterns to help analyze athlete performance and infraction metrics.

2.3 Scope

The scope of the application includes a user-friendly interface for accessing and filtering race walking data stored in an SQL database. The application is able to generate detailed reports and visualizations of the race walking data. It also integrates with external data sources through web scraping.

2.4 Definitions

Bent Knee: Angle of an athlete's leg when the foot is in contact with the ground and the leg's position is before it reaches the vertical position. This infraction symbol is denoted by <

Loss of Contact (LOC): Amount of time when both athlete's feet are not in contact with the ground. This infraction symbol is denoted by ~

Python3: A high-level programming language utilized for driving data pulls, data organization, and visualization of graphs within the Race Walking Data Visualizer.

3. User Interface and Functional Requirements

3.0 - Main Menu: Clicking on each link leads to the specified page. Use the back button to get back to the main menu.

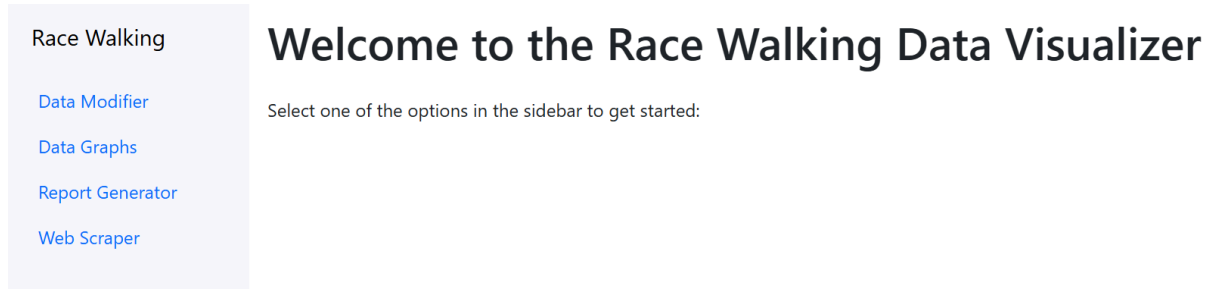


Figure 1 - Main Menu Screen

3.0.1 - Data Table Preview

3.0.2 - Graph Generation

3.0.3 - Report Generator

3.0.4 - Web Scrapping

3.1 Database Interface Requirements

Database :- Allows users to query data and display results in a web interface. Provides a responsive and interactive web UI that enables users to view and edit various database tables.

Database Tables					
Athlete					
AthleteStaging					
Bib					
InfractionSummarySub					
Judge					
JudgeCall					
JudgeInfractionSummary...					
Race					
RaceJudge					
VideoObservation					
VideoObservationPanned					

Table: Athlete					
IDAthlete	LastName	BibNumber	FirstName	CountryCode	Gender
1	A	551	XXXXXX	ESP	F
2	B	163	XXXXXX	PER	F
3	C	135	XXXXXX	GRE	F
4	D	513	XXXXXX	BRA	F
5	E	550	XXXXXX	ESP	F
6	F	164	XXXXXX	PER	F
7	G	151	XXXXXX	JPN	F
8	H	652	XXXXXX	POL	F
9	I	119	XXXXXX	ECU	F
10	J	117	XXXXXX	CZE	F
11	K	109	XXXXXX	CHN	F
12	L	159	XXXXXX	MEX	F
13	M	125	XXXXXX	ESP	F
14	N	614	XXXXXX	JPN	F

Figure 2 - Data Table Preview Screen

3.1.1 Navbar

Display all the necessary tables in the database in alphabetical order. Backup and old tables are hidden. Clicking on a table jumps to the view of that table's data.

3.1.2 Display Tables

3.1.2.1 Visibility The table should only be visible when there are results to display.

3.1.2.2 Editing Users are able to edit a table cell by clicking on it, typing a new value, and clicking off the cell.

3.1.2.3 Column Formatting Each column has a header that describes the data beneath it.

3.1.2.4 Pagination Users navigate long data tables using Previous & Next Buttons to navigate back and forth the table

3.1.2.5 Search Users are able to search for specific values in columns

3.1.2.6 Sorting Users are able to sort each column in alphabetical order or reverse alphabetical order by clicking on column header

3.2 Race Graph Generator

Race Graph Generator :- Provides an interactive UI for selecting a race and bib numbers, then generating a visualization of athlete infractions.

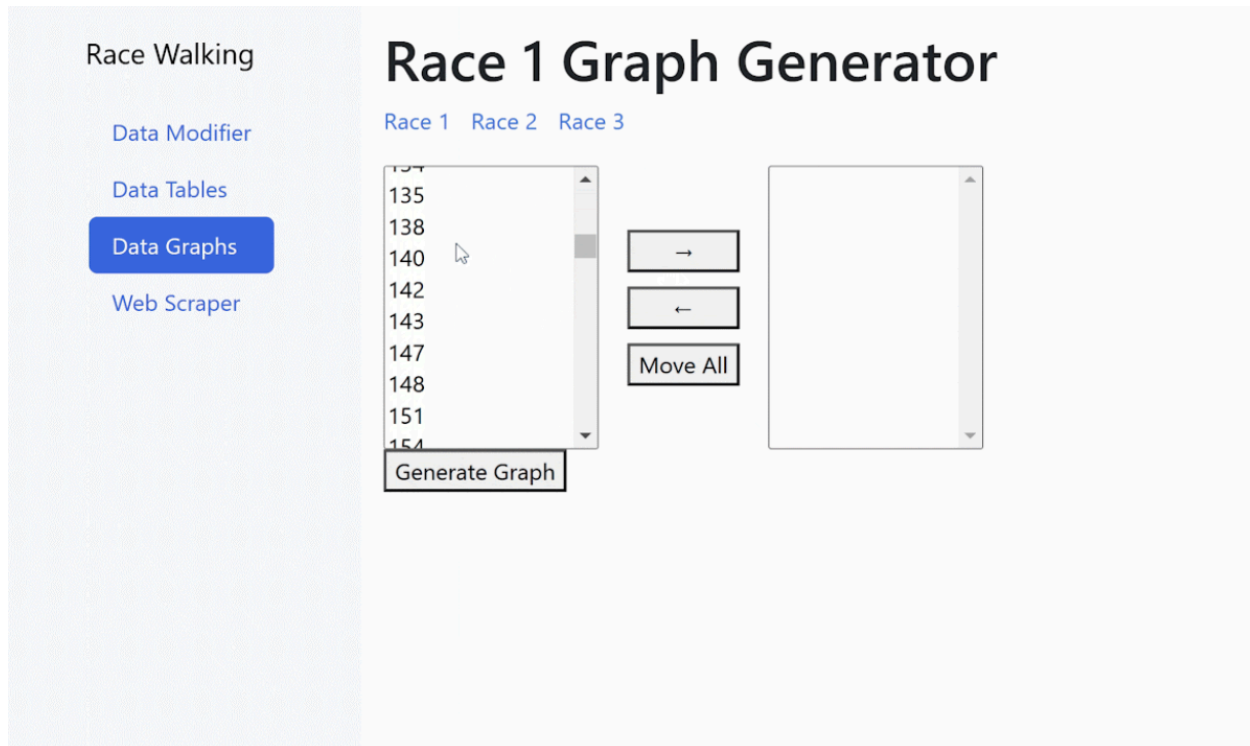


Figure 3 - Race Graph Generator Screen

3.2.1 Race Navigation Users can select which race they want to graph data for one or more walkers.

3.2.2 Generator UI

3.2.2.1 Athlete Selection Users can highlight which athletes they want to generate the graph for in the left panel

3.2.2.1.1 Maximum athletes - Up to 10 athletes per graph.

3.2.2.2 Arrows Users can move the selected athletes to the right panel to generate the graph for that athlete using the right arrow or move any unwanted athletes back to the left side using the left arrow

3.2.2.3 Generate Graph When the user is finishing selecting and moving athletes, they may select this button to generate the graph for the athletes they selected.

3.3 Race Graph Visualization

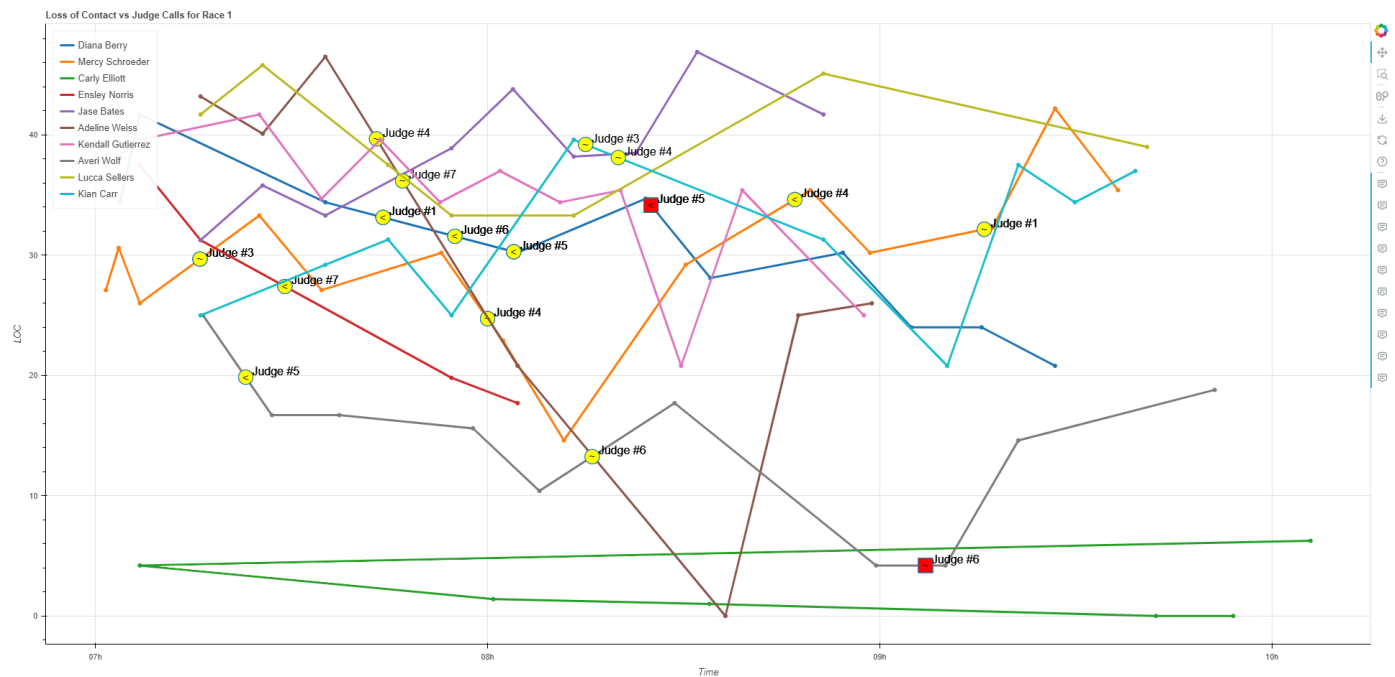


Figure 4 - Loss of Contact vs Judge Calls Graph

3.3.1 Color Scheme Users are able to clearly differentiate between each athlete's plot, using the upper left name table in the graph

3.3.1.1 Color Theory - Utilize color theory to help distinguish one line from another. Use the following ten colors to represent walkers 1 to 10.

- 3.3.1.1.1 Blue - RGB(0, 0, 255)
- 3.3.1.1.2 Orange - RGB(255, 165, 0)
- 3.3.1.1.3 Green - RGB(0, 255, 0)
- 3.3.1.1.4 Red - RGB(255, 0, 0)
- 3.3.1.1.5 Purple - RGB(128, 0, 128)
- 3.3.1.1.6 Brown - RGB(165, 42, 42)
- 3.3.1.1.7 Pink - RGB(255, 192, 203)
- 3.3.1.1.8 Gray - RGB(128, 128, 128)
- 3.3.1.1.9 Yellow-Green - RGB(154, 205, 50)
- 3.3.1.1.10 Cyan - RGB(0, 255, 255)

3.3.2 Markers Users can distinguish between yellow cards and red cards which represent different types of penalties, as well as what type of infraction occurred denoted by either < or ~

3.3.2.1 < - Bent Knee Infraction - The advancing leg must be straightened from the moment of first contact with the ground until the leg is in the vertical upright position. The athlete bent their knee during this period of time.

3.3.2.2 ~ - Loss of Contact Infraction - The athlete visibly appears to have both feet off of the ground at the same time.

3.3.3 Labels Users can see the number of the judge who issued the call for each athlete.

3.3.4 Axes Users can see the x-axis which represents the time and the y-axis which represents the Loss of Contact (LOC). Each access shows the units. The time is represented in seconds (s) and the loss of contact is represented in milliseconds (ms).

3.4 Report Generator

Table Page

Choose a database:
RWComplete.db

Choose a query:
AthleteInfractions.sql

Choose a style:
Striped

Choose a race:
Budapest - 2023-08-24

Download PDF

BibNumber	FirstName	LastName	YellowLOC	YellowBent	RedLOC	RedBent
102	XXXXXX	U	0	4	0	1
106	XXXXXX	D	3	1	0	0
108	XXXXXX	K1	0	2	0	1

Figure 5 - Report Generator Screen

3.4.1 General Page Style and Design

3.4.1.1 Dataset Selector The project has two databases, one for Drexel, and one for a more complete collection of data.

3.4.1.2 Report Selector Should be an intuitive means for the user to select which report they wish to run and produce an output for.

3.4.1.2.1 Number of Infractions Reported by Each Judge

A list of the total number of infractions given in a race.

Including: Yellow Paddle – Loss of Contact, Yellow Paddle – Bent Knee, Red Card – Loss of Contact, Red Card – Bent Knee.

3.4.1.2.2 Correct Number of Red Card Issued per Judge Based on the Video Observations – Totals

List the total number of athletes deserving Red Cards based on video observations. An observation counts as a Red Card for ~ if the walker passes the camera for a progression of steps on two, or more, occasions in which the measured loss of contact is 60 or more ms (milliseconds). An observation counts as a Red Card for < if the walker passes the camera for a progression of steps on two, or more, occasions in which the measured angle at the knee is > 0 degrees and <= 175 degrees.

3.4.1.2.3 Judge Scores A numeric score is determined by comparing the video observations to the recorded infractions on the Summary Sheet.

3.4.1.2.4 Average Loss of Contact by Athlete Across All Observations

This computes the average loss of contact for an athlete across the entire race. This is a bit misleading, because if an athlete has a large loss of

contact at one point and then reduces it, the average may appear legal, but the athlete was still deserving of a red card for loss of contact. Another report addresses this by looking at the two worst laps (over a given threshold like 60ms). This is here mainly for curiosity's sake.

3.4.1.2.5 Average Bent Knee Angle by Athlete Across All Observations Where the Athlete Exhibits a Visible Bend in the Knee

This computes the average bent knee angle for an athlete across the entire race where the athlete exhibits a visible bend in the knee. A visible bend in the knee is defined as > 0 and ≤ 175 degrees. Note, we do not measure the knee angle of walkers who do not violate the bent knee portion of the definition.

3.4.1.2.6 Average Loss of Contact by Athlete for their Two Worst Observations This computes the average loss of contact for the two largest loss of contact measurements for an athlete over the entire race. Only athletes over a threshold (i.e. 60ms) are included in the report.

3.4.1.2.7 Number of Yellow Paddles Issued for an Infraction Without Issuing a Red Card In theory, the judicious use of yellow paddles should help an athlete self-correct and therefore avoid a red card.

3.4.1.2.8 Judge Consistency – Majority

List each judge's name and the number of each infraction that they called that matched the majority of judges. Infractions include Yellow Paddle – Loss of Contact, Yellow Paddle – Bent Knee, Red Card – Loss of Contact, and Red Card – Bent Knee. The majority is defined as half the # of judges (excluding the chief judge).

3.4.1.2.9 Judge Consistency – Discrepancy

List each judge's name and the number of each infraction that they called that did not match the majority of judges. Infractions include Yellow Paddle – Loss of Contact, Yellow Paddle – Bent Knee, Red Card – Loss of Contact, and Red Card – Bent Knee. The majority is defined as half the # of judges (excluding the chief judge).

3.4.1.2.10 Judge Consistency – Missed Calls List each judge's name and the number of each infraction that did not call that the majority of judges did call. Infractions include Yellow Paddle – Loss of Contact, Yellow Paddle – Bent Knee, Red Card – Loss of Contact, and Red Card – Bent Knee. The majority is defined as half the # of judges (excluding the chief judge).

3.4.1.2.11 Athlete Infraction List List each athlete and the number of each type of infraction they received from the judges. Infractions include: Yellow Paddle – Loss of Contact, Yellow Paddle – Bent Knee, Red Card – Loss of Contact, and Red Card – Bent Knee.

3.4.1.2.12 Individual Judge Calls vs Video Observations List each judge and athletes who they gave red cards. List the infraction and then the video observation values for loss of contact and bent knee. If the call is correct, list 'correct,' if the call is incorrect list 'X.'

3.4.1.2.13 Comparison of Static Camera to Panned Camera

List, side by side, the number of infractions (~ and <) for both the static and panned cameras.

3.4.1.2.14 Red Cards without a Yellow Paddle First For each judge, list the number of red cards, for a given infraction, given without having previously given a yellow paddle.

3.4.1.2.15 Legal Athletes with 3 or More Red Cards Athletes that received 3 red cards, but didn't deserve it. Also, include which judges made the incorrect calls and what were the perceived infractions.

3.4.1.3 Style Selector A selector for the user to choose one of many preset styles with which the resulting table will be formatted.

3.4.1.4 Parameter Selector A means of entering extra info needed for the report, which varies depending on the desired report.

3.4.1.4.1 Race Selector The most common extra parameter for a report is the race of interest which, while stored in the database as a number, needs to be displayed to the user in a human readable way. Specifically, a race is displayed by its location and date.

3.4.2 Report Table

3.4.2.1 Headers Contains the information specifying which report this table is for, when it was produced, and with which parameters.

3.2.2.2 Table Contains the resulting report data, styled accordingly.

3.4.3 PDF Functionality

3.4.3.1 Download PDF Button A button which produces and downloads the PDF for the currently produced report.

3.4.3.2 PDF Output Contains headers and labels for report. Table styled accordingly with correct data.

3.4.4 Navigation Bar Navigation bar between this page and the other pages, styled well and intuitively.

3.5 Web Scraper

Athletes

id	name	profile_image_url
1	Álvaro MARTÍN	https://media.aws.iaaf.org/athletes/14410246.jpg

Rankings

id	athlete_id	event_title	ranking_position	ranking_score	weeks_at_position
1	1	Men's 20km Race Walking	1	1372.0	nan
2	1	Men's Overall Ranking	108	1372.0	nan
3	1	Men's 20km Race Walking	1	nan	91.0
4	1	Men's 35km Race Walking	1	nan	61.0

Figure 6: Web Scraper page with Alvaro Martin's data scraped from world athletics.

3.5.1 Overview The web scraper is responsible for gathering race walking data from the worldathletics website. It facilitates automated data acquisition to keep the database up-to-date with the latest results and statistics.

3.5.2 Data Sources

3.5.2.1 Official Race Website The scraper retrieves race results and statistics directly from the worldathletics website;

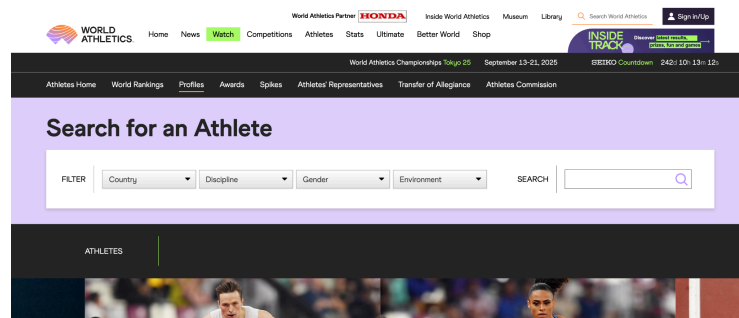


Figure 7: World Athletics - athlete directory homepage:
<https://worldathletics.org/athletes>

3.5.3 Functionalities

3.5.3.1 Automation The scraper can be triggered manually to scrape all data for a race walker athlete.

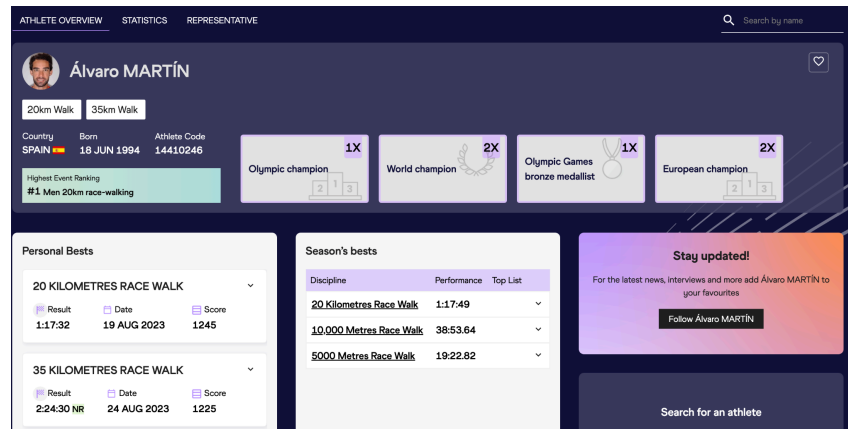


Figure 8: the athlete page for a specific athlete:

<https://worldathletics.org/athletes/spain/alvaro-martin-14410246>

3.5.3.2 Data Parsing Extracts relevant data such as athlete names, times, penalties, and other metrics from web pages.

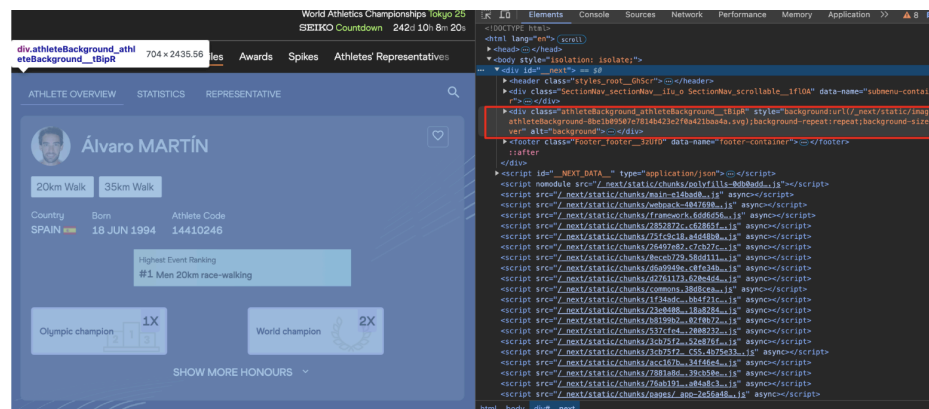


Figure 9: the umbrella div that houses all the website tabs seen in figure 9 for statistics, overview, and representative, as well as all the sub-divs for the desired data mentioned in 3.5.3.2:

(<https://worldathletics.org/athletes/spain/alvaro-martin-14410246>)

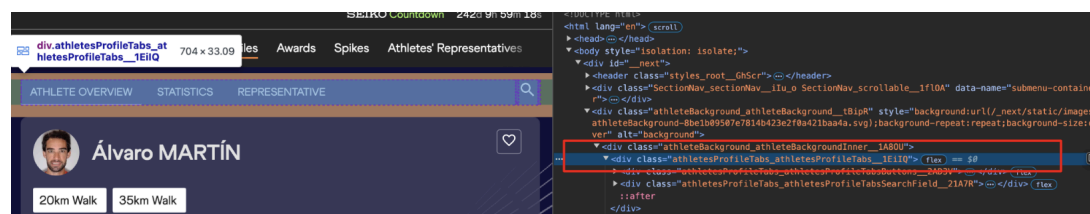


Figure 10: the div that houses the tabs (statistics, overview, and representative) mentioned above, although we are concerned with the statistics and overview tabs

```

<div class="athleteBackground_athleteBackground__tBipR" style="background:url(/_next/static/images/
athleteBackground-8be1b09507e7814b423e2f0a421baa4a.svg);background-repeat:repeat;background-size:co
ver" alt="background">
  <div class="athleteBackground_athleteBackgroundInner__1A80U">
    <div class="athletesProfileTabs_athletesProfileTabs__1EiIQ"> </div> <flex>
    <div class="athletesBio_athletesBio__1Hru0"> </div>
    <div class="athletesOverviewStats_athletesOverviewStats__3UYhF"> <grid>
      <div class="athletesContainer_athletesContainer__3w5eG undefined"> <flex>
        <div class="athletesTitle_athletesTitle__388RT undefined"> <flex == $0
          "Personal Bests"
          ::after
        </div>

```

Figure 11: an example sub-div for the “personal bests” data:

(<https://worldathletics.org/athletes/spain/alvaro-martin-14410246>)

3.5.3.3 Data Storage The collected data is cleaned and stored in the SQL database in a structured format.

3.5.4 Error Handling

3.5.4.1 Resilience Handles connection timeouts, missing data, or changes in the structure of the source websites gracefully.

3.5.4.2 Logging Maintains logs of scraping activity for troubleshooting and auditing purposes.

```

web_scrape.log
1 2024-12-01 20:42:37 - INFO - Successfully clicked the 'Allow all cookies' button.
2 2024-12-01 20:42:37 - INFO - Athlete Name: Álvaro MARTÍN
3 2024-12-01 20:42:37 - INFO - Image URL: https://media.aws.iaaf.org/athletes/14410246.jpg
4 2024-12-01 20:42:37 - INFO - Successfully clicked 'STATISTICS' button.
5 2024-12-01 20:42:37 - INFO - Successfully clicked 'World rankings' button.
6 2024-12-01 20:42:37 - INFO - Extracted ranking data for event #1:
7 2024-12-01 20:42:37 - INFO - Event Title: Men's 20km Race Walking
8 2024-12-01 20:42:37 - INFO - Ranking Position: 1
9 2024-12-01 20:42:37 - INFO - Ranking Score: 1372
10 2024-12-01 20:42:37 - INFO - Weeks at Position: None
11 2024-12-01 20:42:37 - INFO - Extracted ranking data for event #2:
12 2024-12-01 20:42:37 - INFO - Event Title: Men's Overall Ranking
13 2024-12-01 20:42:37 - INFO - Ranking Position: 108
14 2024-12-01 20:42:37 - INFO - Ranking Score: 1372
15 2024-12-01 20:42:37 - INFO - Weeks at Position: None
16 2024-12-01 20:42:37 - INFO - Extracted ranking data for event #3:
17 2024-12-01 20:42:37 - INFO - Event Title: Men's 20km Race Walking
18 2024-12-01 20:42:37 - INFO - Ranking Position: 1
19 2024-12-01 20:42:37 - INFO - Ranking Score: None

```

Figure 12: a piece of the log file from the webscraper.py program (web_scrape.log)

4. Nonfunctional Requirements

4.1 Human Factors

4.1.1 Types of Users

4.1.1.1 Users who besides simply generating reports and graphs, will update the data using both the table modification page, and the webscraper.

4.1.1.2 Users that simply will generate reports and graphs from the existing data.

4.2 Documentation

4.2.1 Interactive Visualization Documentation

4.2.1.1 Documentation is provided on the use of the produced visualizations.

4.2.2 Application Architecture Documentation

4.2.2.1 Documentation is provided on the architecture of the system in order to specify how the functional requirements are implemented.

4.3 Hardware

4.3.1 Computer The client must have a computer with the following hardware:

4.3.1.1 A keyboard

4.3.1.2 A pointing device

4.3.1.3 A monitor

4.3.2 Network The client's computer need not have an internet connection, unless they are using the webscraping functionality.

4.3.3 Download Capability Users must be able to download the generated graphs directly from the web interface.

4.4 Software

4.4.1 Operating System Recommended System Latest versions of Windows, macOS, or stable Linux distributions like Ubuntu 20.04 LTS.

4.4.2 Python Version Required Python 3.9 or higher.

4.4.3 Libraries and Frameworks Flask, Pandas, SQLite3, Bokeh, Selenium, Headless Chrome and NumPy.

4.4.4 Development Tools Visual Studio Code, PyCharm, Git.

4.4.5 Web Browser Google Chrome, Firefox, Safari, Edge (latest versions).

4.4.6 Database Development SQLite

4.5 Performance

4.5.1 Response Time

The response times listed below are designed to ensure that the application remains responsive under normal operational conditions and within the typical workload parameters.

4.5.1.1 User Interaction

User interactions with the system that do not require loading data from the server or off-client communication should have a response time of under 0.5 seconds.

4.5.1.2 Data Fetching and Display

The system response time for fetching data based on user queries and displaying it in the Dash interface should be under 2 seconds.

4.5.1.3 Graph Generation and Download

The response time from the initiation of graph generation to the availability of the file for download should not exceed 10 seconds. The actual download time of the graph image file may vary depending on the user's internet connection speed.

4.5.1.4 System Load Handling

The system should handle up to 50 concurrent users without significant degradation in response times specified above.

4.5.1.5 Database Operations

Database queries, whether for fetching data for the Flask app or for generating graphs, should respond within 3 seconds under typical usage conditions.

4.6 Error Handling and Reliability

4.6.1 Input Errors

Handling of input errors is crucial for maintaining data integrity and providing a user-friendly experience.

4.6.1.1 Error Messaging

When an input error is detected, the system must immediately display a contextual error message to the user.

4.6.1.2 User Acknowledgment

Users must acknowledge any error messages relating to input errors before they can continue to use the application.

4.7 Security

4.7.1 Application Access

Access to the application is restricted to users within the organizational network or those authenticated.

4.7.2 Database Security

Database access is limited to system administrators and specifically authorized personnel.

4.8 Usability of code

Python is a cross-platform language which is able to run in most popular systems such as Windows, MacOS, and Linux. As long as the correct version of python and the required libraries are installed, any systems are able to run the code. As a requirement, the requirements file, which allows a user to quickly install all the required libraries with their correct versions, ought to be kept up to date, for ease of use.

5. System Evolution

5.1 New widgets:

As the Race Walking Data Visualizer application evolves, the integration of new widgets will be crucial to enhancing user experience and expanding functionality. These widgets will cater to diverse user needs and enable more comprehensive analysis of race walking data. Some potential new widgets include:

5.1.1 Interactive Graphs Introduce interactive graphing capabilities to allow users to visualize race walking data dynamically. Graphs such as scatter plots, histograms, and line charts can provide deeper insights into athlete performance and referee calls.

5.1.2 Custom Filters Implement custom filter options to empower users to refine data queries based on specific criteria. Custom filters could include parameters related to athlete demographics, race conditions, or referee observations, enhancing the precision and relevance of data analysis.

5.1.3 Comparison Tools Develop tools for comparing multiple races, athletes, or referee judgments side by side. Comparative analysis features can facilitate benchmarking, trend identification, and performance evaluation across different events or time periods.

5.1.4 Data Export Formats Expand the range of export formats beyond PDFs to accommodate diverse user preferences and downstream analysis requirements. Additional export options such as Excel spreadsheets, or CSV files will enhance the versatility and usability of the application.

5.1.5 Combined Report Page Right now there are report types that are simply database queries. In the future, could possibly have reports that combine all the elements we have so far made. A report schema that can be run with given parameters that includes not just a table, but a specific series of tables, and not just tables, but tables and graphs, all in the PDF would be relatively simple to do. Could just make some preset 'composite reports' that are constituted from the existing report queries and also specific graph specifications.

5.2 User Permissions

As the Race Walking Data Visualizer evolves, the management of user permissions will become increasingly important to ensure data security, integrity, and accessibility. Implementing robust user permission controls will enable administrators to define and enforce access levels, safeguard sensitive information, and maintain regulatory compliance. Presently, database access is limited to system administrators and specifically authorized personnel.